
Lab 1 Practice

Procedural Language/Structured Query Language

CSE 4508
RELATIONAL DATABASE MANAGEMENT SYSTEM LAB

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1 PL/SQL

PL/SQL is a block-structured language where we can write code organized into blocks similar to Java/C/C++, although the coding style differs here due to not having any curly braces to define blocks, rather we define blocks using the BEGIN and END keywords. You can define PROCEDURES (which execute some code without returning anything) and FUNCTIONS (which execute code and return some variable/record). And you can even define unnamed blocks in PL/SQL, otherwise known as anonymous blocks that can be called immediately upon defining using the / at the end.

1.1 Structure of PL/SQL Block

PL/SQL extends SQL by adding constructs found in procedural languages, resulting in a structural language that is more powerful than SQL. The basic unit in PL/SQL is a block. All PL/SQL programs are made up of blocks, which can be nested within each other.

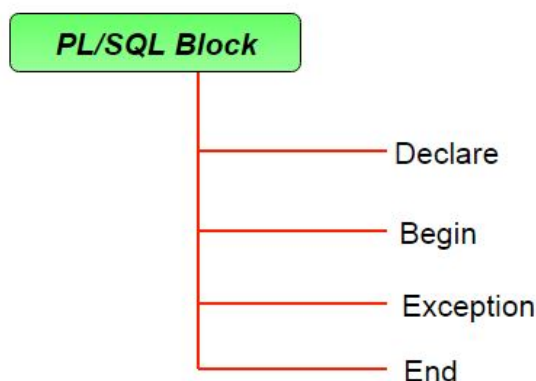


Figure 1: Structure of PL/SQL Block

1.2 Syntax of PL/SQL Block

```
DECLARE
    declaration statements;

BEGIN
    executable statements;

EXCEPTION
    exception handling statements;

END;
/
```

Note: Remember to SET SERVEROUTPUT ON to see the results of the blocks.

```
SQL> SET SERVEROUTPUT ON;
SQL> BEGIN
  2   DBMS_OUTPUT.PUT_LINE('Hello World');
  3 END;
  4 /
Hello World

PL/SQL procedure successfully completed.
```

Figure 2: PL/SQL Syntax to show output

1.3 Example of PL/SQL Block

```
DECLARE
    v_salary NUMBER;
    v_bonus  NUMBER := 1000;

BEGIN
    SELECT salary INTO v_salary FROM employees WHERE employee_id = 101;

    IF v_salary > 5000 THEN
        v_salary := v_salary + v_bonus;
    END IF;

    DBMS_OUTPUT.PUT_LINE('Updated salary: ' || v_salary);

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Employee not found.');
```

```
    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE('An error occurred.');
```

```
END;
/
```

2 Function

A function in PL/SQL is a subprogram designed to perform a specific task and return a single value. Functions are commonly used when a value needs to be computed and returned, such as the result of a calculation, string manipulation, or database query. Unlike procedures, a function must always return a value.

2.1 Example of a Function

Define a function that, given the name of a department, returns the count of the number of instructors in that department.

```
CREATE FUNCTION dept_count (dept_name VARCHAR(20))
RETURNS INTEGER
BEGIN
    DECLARE d_count INTEGER;
    SELECT COUNT(*)
    INTO d_count
    FROM instructor
    WHERE instructor.dept_name = dept_name;
    RETURN d_count;
END;
/
```

Running the function:

```
SELECT dept_name, budget
FROM department
WHERE dept_count(dept_name) > 12;
```

2.2 Table Functions

```
CREATE FUNCTION instructor_of (dept_name CHAR(20))
RETURNS TABLE (
```

```
    ID VARCHAR(5),
    name VARCHAR(20),
    dept_name VARCHAR(20),
    salary NUMERIC(8,2))
RETURN TABLE
    (SELECT ID, name, dept_name, salary
     FROM instructor
     WHERE instructor.dept_name = dept_name);
/
```

Running the function:

```
SELECT *
FROM TABLE (instructor_of('Music'));
```

3 Procedure

A procedure in PL/SQL is also a subprogram. It can accept input and output parameters. It uses parameters to return results, but it does not use the RETURN statement. It is used to perform an action (eg. modifying data in the database).

3.1 Example of a Procedure

```
CREATE PROCEDURE dept_count_proc (dept_name IN VARCHAR(20), d_count OUT
    INTEGER)
BEGIN
    SELECT COUNT(*) INTO d_count
    FROM instructor
    WHERE instructor.dept_name = dept_count_proc.dept_name;
END;
/
```

Running the procedure:

```
DECLARE d_count INTEGER;
CALL dept_count_proc('Physics', d_count);
```

4 Language Constructs

PL/SQL syntax for conditionals, loops, functions, and mathematical operations.

4.1 Conditionals

```
DECLARE
    num NUMBER := 7;
BEGIN
    IF MOD(num, 2) = 0 THEN
        DBMS_OUTPUT.PUT_LINE('The number is even.');
```

```
    ELSE
        DBMS_OUTPUT.PUT_LINE('The number is odd.');
```

```
    END IF;
END;
/
```

4.2 Loops

Print numbers from 1 to 5 using a WHILE loop.

```
DECLARE
    counter NUMBER := 1;
```

```
BEGIN
    WHILE counter <= 5 LOOP
        DBMS_OUTPUT.PUT_LINE('Number: ' || counter);
        counter := counter + 1;
    END LOOP;
END;
/
```

Print numbers from 1 to 5 using a FOR loop

```
BEGIN
    FOR i IN 1..5 LOOP
        DBMS_OUTPUT.PUT_LINE('Number: ' || i);
    END LOOP;
END;
/
```

4.3 Function inside a Procedure

```
CREATE OR REPLACE FUNCTION calculate_square(n NUMBER) RETURN NUMBER IS
BEGIN
    RETURN n * n;
END calculate_square;
/

CREATE OR REPLACE PROCEDURE print_square(num IN NUMBER) IS
    result NUMBER;
BEGIN
    result := calculate_square(num);
    DBMS_OUTPUT.PUT_LINE('The square of ' || num || ' is: ' || result);
END print_square;
/

BEGIN
    print_square(5);
END;
/
```

4.4 Mathematical Functions

```
DECLARE
    num NUMBER := 16;
    result NUMBER;
BEGIN
    result := SQRT(num);
    DBMS_OUTPUT.PUT_LINE('The square root of ' || num || ' is ' || result);
END;
/
```

```
DECLARE
    num1 NUMBER := 10;
    num2 NUMBER := 3;
BEGIN
    DBMS_OUTPUT.PUT_LINE('The remainder when ' || num1 || ' is divided by ' || num2 || ' is ' || MOD(num1, num2));
END;
/
```

5 Tasks

Task 1

A hacker stumbles onto a database table containing only two columns:

- Username (varchar2) and
- Password_Length (number).

Password_Length only contains a number, such as 7 or 8, denoting how long the password of that username is.

If necessary, write a block of PL/SQL, using a function that will first find the highest Password_Length from the table. It will then determine how many permutations the hacker must complete to crack that password.

The password only contains letters of the alphabet. You therefore have $26 \times 2 = 52$ possibilities for each symbol of the password. However, no character can be repeated. So the correct answer, for a password of length 4, is:

$$52 \times 51 \times 50 \times 49 = 6,497,400$$

Task 2

Write a PL/SQL procedure that takes as input a string. The program will achieve two things:

1. Make a new string with a space added between every character of the input string. For instance, if the input string is “racecar”, the output will be “r a c e c a r”.
2. Check if the original input string was a palindrome. Print “Yes” or “No” accordingly. (For example: ‘racecar’ is a palindrome, but ‘tracecar’ is not.)

Task 3

A simple library system needs to be created to track the borrowing of books. For each book borrowed, you need to check when it is due and calculate any late fees.

Task 3.1

Create a Library_Borrowing table with the following columns:

- B_ID (number),
- Name (varchar2),
- Book_Title (varchar2),
- Borrow_Date (date),
- Due_Date (date),
- Returned_Date (date, nullable).

Task 3.2

Insert the following dummy borrowing records into the Library_Borrowing table:

B_ID	Name	Book_Title	Borrow_Date	Due_Date	Returned_Date
1	John Doe	The Great Gatsby	2024-08-01	2024-08-15	2024-08-20
2	Jane Smith	1984	2024-08-10	2024-08-24	NULL
3	Alice Johnson	To Kill a Mockingbird	2024-09-01	2024-09-15	2024-09-16
4	Bob Brown	Moby Dick	2024-08-20	2024-09-03	2024-09-02
5	Charlie Adams	The Catcher in the Rye	2024-09-05	2024-09-19	NULL

Task 3.3

Write a PL/SQL function `Calculate_Late_Fee` that calculates the late fee for a borrowing record. The late fee is calculated as \$1.00 per day after the `Due_Date`. The function should return the total late fee for the record. If the book has not been returned (i.e., `Returned_Date` is NULL), calculate the fee as of the current date.

Task 3.4

Write a PL/SQL procedure `List_Overdue_Books` that lists all members who have not yet returned their books, along with the number of days their books are overdue, if any. If the current date is past the `Due_Date`, the book is considered overdue.

For example:

- Jane Smith has not yet returned the book "1984". If today's date is 2024-09-18, the book is overdue by 25 days.
- Charlie Adams has not yet returned the book "The Catcher in the Rye". If today's date is 2024-09-18, the book is not yet overdue, but it is due soon.